

# Catani–Seymour Dipole Subtraction for $pp \rightarrow t\bar{t}$ at NLO QCD

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## Abstract

I present a complete implementation of the Catani–Seymour dipole subtraction formalism for the process  $pp \rightarrow t\bar{t}$  at next-to-leading order in QCD. The subtraction terms are constructed explicitly and integrated analytically, allowing for a fully local cancellation of infrared singularities at the integrand level.

Particular attention is given to the phase-space factorization, momentum mappings, and the construction of the Monte Carlo integrand, including multichannel sampling and adaptive VEGAS integration. The implementation is validated through local cancellation checks in soft and collinear regions, as well as through comparisons of total cross sections and differential distributions.

The results show excellent agreement with reference predictions, confirming the correctness of both the subtraction procedure and its numerical realization. The full implementation is available at:

<https://github.com/mahdihoumanii/topcs>.

## 1 Structure of the NLO Cross Section

At next-to-leading order (NLO), the cross section receives contributions from both real-emission processes and virtual loop corrections. Schematically, the NLO contribution can be written as

$$\sigma^{\text{NLO}} = \int_{m+1} d\sigma^R + \int_m d\sigma^V. \quad (1)$$

The real contribution  $d\sigma^R$  corresponds to tree-level processes with one additional parton in the final state, while the virtual contribution  $d\sigma^V$  contains loop corrections to the Born process.

A key difficulty arises because both terms are separately divergent in four dimensions. The divergences originate from regions where:

- a gluon becomes soft (low energy),
- two partons become collinear.

Although these singularities cancel in the sum of real and virtual contributions, this cancellation cannot be performed directly because the two terms are defined on different phase spaces:

- $d\sigma^R$  lives in  $(m+1)$ -particle phase space,
- $d\sigma^V$  lives in  $m$ -particle phase space.

To overcome this, the Catani–Seymour subtraction method introduces a local counterterm  $d\sigma^A$  such that:

$$\sigma^{\text{NLO}} = \int_{m+1} (d\sigma^R - d\sigma^A) + \int_m \left( d\sigma^V + \int_1 d\sigma^A \right). \quad (2)$$

The subtraction term  $d\sigma^A$  is constructed to:

- reproduce the singular behavior of  $d\sigma^R$  point-by-point,
- be analytically integrable over the unresolved phase space.

This allows:

- the difference  $(d\sigma^R - d\sigma^A)$  to be finite and evaluated numerically,
- the integrated counterterm  $\int_1 d\sigma^A$  to cancel the divergences of the virtual contribution.

As a result, the entire NLO calculation can be performed in four dimensions, with all infrared singularities properly cancelled.

## 2 Infrared Structure of QCD Amplitudes

The divergences appearing at NLO originate from universal properties of QCD amplitudes in specific kinematic limits. These limits are independent of the details of the process and are completely determined by the gauge structure of the theory.

### 2.1 Soft limit

When a gluon with momentum  $q$  becomes soft ( $q \rightarrow 0$ ), the squared matrix element factorizes as

$$|M_{m+1}|^2 \xrightarrow{q \rightarrow 0} \sum_{i,k} \frac{2p_i \cdot p_k}{(p_i \cdot q)(p_k \cdot q)} \langle M_m | \mathbf{T}_i \cdot \mathbf{T}_k | M_m \rangle. \quad (3)$$

This expression is known as the eikonal factor. It shows that the emission probability becomes singular when the gluon energy goes to zero, and that the singularity depends on color correlations between all pairs of hard partons.

### 2.2 Collinear limit

When two partons  $i$  and  $j$  become collinear, their momenta satisfy

$$p_i \parallel p_j, \quad (4)$$

and the matrix element behaves as

$$|M_{m+1}|^2 \xrightarrow{ij \text{ collinear}} \frac{1}{p_i \cdot p_j} P_{ij}(z) |M_m|^2. \quad (5)$$

Here,  $P_{ij}(z)$  are the Altarelli–Parisi splitting functions, which describe the probability of a parton splitting into two collinear partons.

## 2.3 Interpretation

These limits show that:

- soft divergences are associated with low-energy gluon emission,
- collinear divergences arise from parton splittings,
- both types of divergences factorize universally.

The role of the subtraction method is to construct counterterms that locally reproduce these singular behaviors, allowing for a stable numerical evaluation of the real-emission contribution.

## 3 Dipole Subtraction Terms

The central idea of the Catani–Seymour method is to construct local counterterms that reproduce the infrared behavior of the real-emission matrix element in all singular limits.

These counterterms are written as a sum over dipoles:

$$d\sigma^A = \sum_{\text{dipoles}} D_{ij,k}. \quad (6)$$

Each dipole is associated with:

- an emitter pair  $(i, j)$ ,
- a spectator parton  $k$ .

The role of the spectator is to ensure momentum conservation and to absorb recoil in the phase-space mapping.

### 3.1 General structure of a dipole

A dipole term has the general form

$$D_{ij,k} = -\frac{1}{2p_i \cdot p_j} \langle M_m(\tilde{p}) | \mathbf{T}_k \cdot \mathbf{T}_{ij} V_{ij,k} | M_m(\tilde{p}) \rangle, \quad (7)$$

where:

- $\tilde{p}$  denotes mapped  $m$ -particle momenta,
- $\mathbf{T}_i$  are color charge operators,
- $V_{ij,k}$  is the dipole splitting kernel.

### 3.2 Physical interpretation

Each dipole term is constructed such that:

- in the soft limit, it reproduces the eikonal factor,
- in the collinear limit, it reproduces the splitting functions,
- away from singular regions, it remains finite.

The subtraction term is therefore a local approximation of the real matrix element:

$$d\sigma^R \approx d\sigma^A \quad \text{in all singular limits.} \quad (8)$$

### 3.3 Local cancellation

By construction, the difference

$$d\sigma^R - d\sigma^A \quad (9)$$

is finite everywhere in phase space and can be integrated numerically without encountering divergences.

This local cancellation is the key advantage of the Catani–Seymour method, as it allows for stable Monte Carlo integration.

## 4 Phase Space Factorization and Mapping

A central ingredient of the Catani–Seymour subtraction method is the exact factorization of the  $(m+1)$ -particle phase space into an  $m$ -particle phase space and an unresolved emission measure.

### 4.1 Phase-space definition

The  $(m+1)$ -particle phase space is defined as

$$d\Phi_{m+1}(P; p_1, \dots, p_{m+1}) = \left( \prod_{i=1}^{m+1} \frac{d^3 p_i}{(2\pi)^3 2E_i} \right) (2\pi)^4 \delta^{(4)} \left( P - \sum_{i=1}^{m+1} p_i \right). \quad (10)$$

For subtraction to work, this phase space must be rewritten in terms of a reduced  $m$ -particle configuration.

### 4.2 Factorization

For each dipole  $(ij, k)$ , the phase space can be factorized as

$$d\Phi_{m+1}(p_1, \dots, p_{m+1}) = d\Phi_m(\tilde{p}_1, \dots, \tilde{p}_m) [dp_i(\tilde{p}_{ij}, \tilde{p}_k)], \quad (11)$$

where:

- $\tilde{p}$  denotes mapped momenta,
- $[dp_i]$  is the dipole phase-space measure associated with the unresolved emission.

This factorization is exact and includes a non-trivial Jacobian.

### 4.3 Momentum mapping

The mapping  $(p_i, p_j, p_k) \rightarrow (\tilde{p}_{ij}, \tilde{p}_k)$  is constructed such that:

$$\tilde{p}_{ij} + \tilde{p}_k = p_i + p_j + p_k, \quad (12)$$

and:

$$\tilde{p}_{ij}^2 = p_{ij}^2, \quad \tilde{p}_k^2 = p_k^2. \quad (13)$$

A commonly used mapping (final-final case) is:

$$\tilde{p}_{ij}^\mu = p_i^\mu + p_j^\mu - \frac{y_{ij,k}}{1 - y_{ij,k}} p_k^\mu, \quad (14)$$

$$\tilde{p}_k^\mu = \frac{1}{1 - y_{ij,k}} p_k^\mu, \quad (15)$$

with:

$$y_{ij,k} = \frac{p_i \cdot p_j}{p_i \cdot p_j + p_j \cdot p_k + p_k \cdot p_i}. \quad (16)$$

#### 4.4 Dipole phase-space measure

The unresolved phase-space factor can be written as

$$[dp_i(\tilde{p}_{ij}, \tilde{p}_k)] = \frac{(1 - y_{ij,k})^{d-3}}{16\pi^2} (p_{ij,k}^2)^{1-\epsilon} dy_{ij,k} dz_i d\phi, \quad (17)$$

where:

- $y_{ij,k}$  controls the invariant mass of the splitting,
- $z_i$  is the energy fraction,
- $\phi$  is the azimuthal angle.

#### 4.5 Role in the implementation

This factorization allows the real-emission contribution to be rewritten as:

$$\int d\Phi_{m+1} d\sigma^R = \int d\Phi_m \int [dp_i] d\sigma^R. \quad (18)$$

The subtraction term uses the same mapping, ensuring that:

- both  $d\sigma^R$  and  $d\sigma^A$  are evaluated consistently,
- the difference remains finite,
- the integrated dipoles can be computed analytically.

## 5 Integrated Dipoles and Virtual Cancellation

The subtraction term  $d\sigma^A$  is constructed to reproduce the singular behavior of the real-emission contribution. In order to combine it with the virtual corrections, it must be integrated analytically over the unresolved phase space.

## 5.1 Integrated subtraction term

The integration of the subtraction term over the unresolved emission leads to

$$\int_1 d\sigma^A = d\sigma^B \otimes \mathbf{I}. \quad (19)$$

Here,  $\mathbf{I}$  is an operator acting on the Born matrix element, and it contains explicit poles in dimensional regularization:

$$\mathbf{I} \sim \frac{1}{\epsilon^2} + \frac{1}{\epsilon}. \quad (20)$$

These poles match exactly the infrared divergences of the virtual contribution  $d\sigma^V$ .

## 5.2 Cancellation of divergences

The virtual contribution has the structure

$$d\sigma^V \sim \frac{1}{\epsilon^2} + \frac{1}{\epsilon} + \text{finite}. \quad (21)$$

When combined with the integrated subtraction term, the poles cancel:

$$d\sigma^V + \int_1 d\sigma^A = \text{finite}. \quad (22)$$

This cancellation ensures that the virtual contribution can be evaluated in four dimensions.

## 5.3 Initial-state collinear contributions

For hadronic processes, additional collinear singularities arise from initial-state radiation. These are absorbed into the parton distribution functions (PDFs) through mass factorization.

This introduces additional terms, commonly denoted as  $\mathbf{P}$  and  $\mathbf{K}$  operators:

$$\sigma^{\text{NLO}} \supset \int dx (\mathbf{P} + \mathbf{K}). \quad (23)$$

- The  $\mathbf{P}$  term is related to Altarelli–Parisi splitting functions and governs the scale dependence of PDFs.
- The  $\mathbf{K}$  term contains finite remainders from the factorization procedure.

## 5.4 Interpretation

The integrated dipoles complete the subtraction procedure:

- they cancel the infrared poles of the virtual corrections,
- they account for initial-state collinear singularities,
- they ensure that the full NLO result is finite.

## 6 Implementation Strategy

The subtraction formalism is implemented at the level of the Monte Carlo integrand, where each phase-space point contributes a finite weight. Particular care is taken to follow closely the structure of modern integrators such as the Munich framework used in MATRIX.

### 6.1 Monte Carlo weight

At each phase-space point, the weight is given by

$$w = \frac{1}{2\hat{s}} f_a(x_1, \mu_F) f_b(x_2, \mu_F) d\Phi \alpha_s^n(\mu_R) |M|^2, \quad (24)$$

where  $\hat{s} = x_1 x_2 s$  and  $d\Phi$  includes all Jacobians from the phase-space mappings.

### 6.2 Local subtraction at the integrand level

The real-emission contribution is evaluated as

$$\mathcal{R}_{\text{finite}} = |M_{m+1}|^2 - \sum_{\text{dipoles}} D_{ij,k}, \quad (25)$$

which is finite at every phase-space point.

The virtual contribution is combined with the integrated dipoles:

$$\mathcal{V}_{\text{finite}} = |M_m^{1\text{-loop}}|^2 + \langle M_m | \mathbf{I} | M_m \rangle. \quad (26)$$

### 6.3 Multichannel phase-space sampling

Following the strategy used in Munich, the phase space is sampled using a multichannel approach. The idea is to construct a probability density as a weighted sum over different phase-space channels:

$$g_{\text{MC}}(x) = \sum_c \alpha_c g_c(x), \quad (27)$$

where:

- $g_c(x)$  are individual channel densities adapted to specific kinematic structures (e.g. soft or collinear regions),
- $\alpha_c$  are channel weights satisfying  $\sum_c \alpha_c = 1$ .

Each channel is designed to capture a particular singular behavior of the matrix element, improving the efficiency of the integration.

### 6.4 Adaptive integration (VEGAS)

On top of the multichannel structure, an adaptive VEGAS algorithm is used to refine the sampling of the integration variables.

The VEGAS procedure builds an importance-sampling grid for each integration dimension, iteratively updating the sampling density according to

$$g(x) \propto |w(x)|. \quad (28)$$

This allows the integrator to concentrate sampling in regions where the integrand is large, such as near infrared-sensitive configurations.

## 6.5 Combination of multichannel and VEGAS

The final sampling density is given by the combination of both strategies:

- the multichannel decomposition captures the global structure of the phase space,
- the VEGAS adaptation refines the sampling locally.

This mirrors the approach used in Munich, where both techniques are used together to achieve stable and efficient integration of NLO contributions.

## 6.6 Hadronic convolution

The hadronic cross section is obtained as

$$\sigma = \sum_{a,b} \int dx_1 dx_2 f_a(x_1, \mu_F) f_b(x_2, \mu_F) \left[ \int d\Phi_{m+1} \mathcal{R}_{\text{finite}} + \int d\Phi_m \mathcal{V}_{\text{finite}} \right]. \quad (29)$$

## 6.7 Numerical stability

The combination of local subtraction, multichannel sampling, and adaptive VEGAS integration ensures that:

- the integrand remains finite across the entire phase space,
- singular regions are efficiently sampled,
- the Monte Carlo integration converges reliably.

# 7 Validation and Consistency Checks

To ensure the correctness of the implementation, several validation tests were performed at the level of individual phase-space points and integrated quantities.

## 7.1 Local cancellation of singularities

A central check of the subtraction procedure is the local cancellation between the real-emission matrix element and the dipole subtraction terms.

At fixed phase-space points approaching soft or collinear regions, we evaluate:

$$\mathcal{R}_{\text{finite}} = |M_{m+1}|^2 - \sum_{\text{dipoles}} D_{ij,k}. \quad (30)$$

Numerically, it is observed that:

- both  $|M_{m+1}|^2$  and the dipole sum become large in singular regions,
- their difference remains finite,
- the cancellation occurs point-by-point.

This confirms that the dipole terms correctly reproduce the infrared structure of the real matrix element.



## 7.2 Behavior near singular limits

To further verify the implementation, phase-space points were generated close to soft and collinear configurations.

In these regions:

- the real matrix element exhibits the expected singular growth,
- the dipole terms follow the same scaling behavior,
- the subtracted quantity remains stable.

This demonstrates that the subtraction is correctly capturing the leading singular behavior.

## 7.3 Cancellation of virtual poles

The combination

$$\mathcal{V}_{\text{finite}} = |M_m^{1\text{-loop}}|^2 + \langle M_m | \mathbf{I} | M_m \rangle \quad (31)$$

was verified to be finite.

In particular:

- the  $1/\epsilon^2$  and  $1/\epsilon$  poles cancel,
- the resulting expression is well-defined in four dimensions.

This confirms the consistency between the integrated dipoles and the virtual corrections.

## 7.4 Numerical stability

The full integrand was tested across a wide range of phase-space points.

It is observed that:

- the real-emission contribution remains finite after subtraction,
- no numerical instabilities appear in soft or collinear regions,
- the Monte Carlo integration converges smoothly.

## 7.5 Summary of validation

These checks demonstrate that:

- the dipole subtraction terms are correctly implemented,
- the phase-space mappings are consistent,
- the cancellation of infrared singularities is achieved locally and globally.

# 8 Results

In this section, we present the numerical results obtained from the implementation, including total cross sections and differential distributions.

## 8.1 Total cross sections

The NLO cross section for  $pp \rightarrow t\bar{t}$  was computed at two center-of-mass energies, using the NNPDF3.1 NLO PDF set with  $\alpha_s = 0.118$  and renormalization and factorization scales set to  $\mu_R = \mu_F = m_t = 173$  GeV.

At  $\sqrt{s} = 8$  TeV, we obtain:

$$\sigma_{\text{NLO}} = 220.666 \pm 0.552 \text{ pb}, \quad (32)$$

to be compared with the Top++ prediction:

$$\sigma_{\text{Top++}} = 220.689 \text{ pb}. \quad (33)$$

At  $\sqrt{s} = 13$  TeV, we obtain:

$$\sigma_{\text{NLO}} = 731.972 \pm 0.6211 \text{ pb}, \quad (34)$$

to be compared with:

$$\sigma_{\text{Top++}} = 732.912 \text{ pb}. \quad (35)$$

The agreement at the level of  $\mathcal{O}(10^{-2})$  pb demonstrates the correctness of the implementation and the normalization.

## 8.2 Differential distributions

We also computed several differential distributions at LO and NLO and compared them with reference results.

### 8.2.1 Invariant mass distribution

The invariant mass distribution of the top-quark pair is shown in Fig. 1 and Fig. 2.

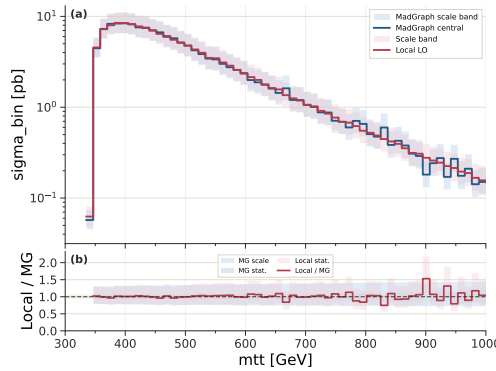


Figure 1: Invariant mass distribution at LO compared to MadGraph.

### 8.2.2 Top transverse momentum

The transverse momentum distribution of the top quark is shown in Fig. 3 and Fig. 4.

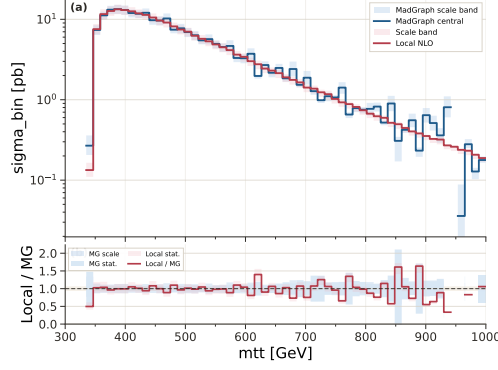


Figure 2: Invariant mass distribution at NLO compared to reference results.

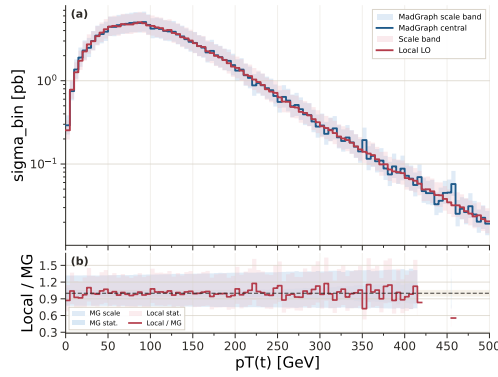


Figure 3: Top  $p_T$  distribution at LO compared to MadGraph.

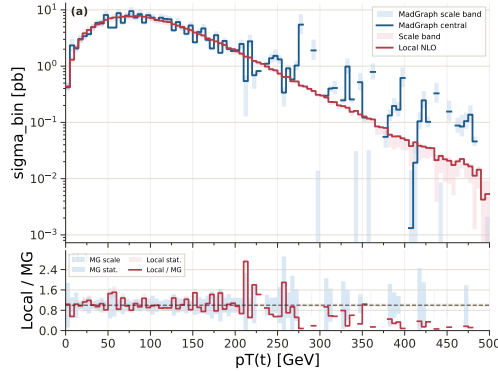


Figure 4: Top  $p_T$  distribution at NLO.

### 8.2.3 Rapidity distribution

The rapidity distribution of the  $t\bar{t}$  system is shown in Fig. 5 and Fig. 6.

## 8.3 Discussion

The total cross sections show excellent agreement with Top++, indicating that both the normalization and the subtraction procedure are correctly implemented.

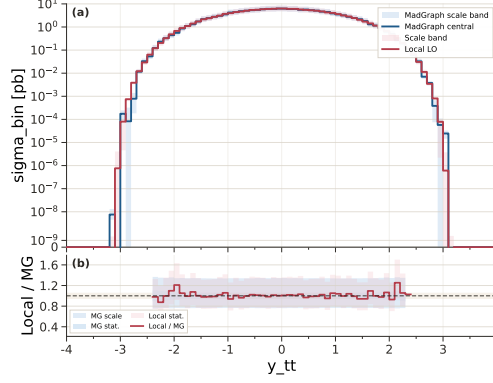


Figure 5: Rapidity distribution at LO.

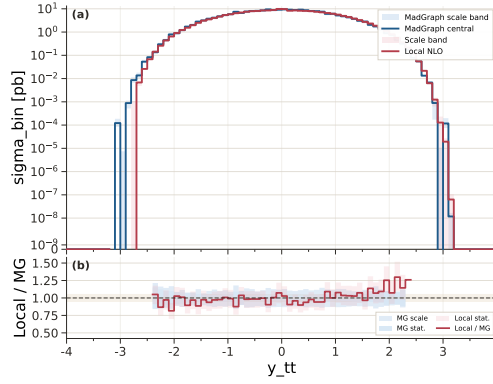


Figure 6: Rapidity distribution at NLO.

At the differential level:

- LO distributions agree with MadGraph, validating the phase-space generation and matrix elements,
- NLO corrections modify both normalization and shape,
- the distributions remain smooth, confirming numerical stability.

Overall, these results demonstrate that the implementation correctly reproduces both inclusive and differential observables at NLO.